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Root Cause Analysis - 2-RV-102 Elbow Leak

Scope and Responsibilities

This document details the root cause analysis following the pinhole leak identified on 2-RV-102 within the ATU-100. The investigation focuses on the failure of a 90° elbow in the regenerator overhead circuit, which resulted in an unscheduled isolation and clamp installation on 2007-09-14. The analysis evaluates the effectiveness of the existing inspection strategy and the localized corrosion mechanisms present in acid gas service.

Douglas Ferreira, as Inspection Supervisor and API 570 inspector, is responsible for the technical review of NDE data and the assessment of corrosion rates against site retirement thickness limits. Gordon Lisle, as Unit Engineer, is responsible for the engineering evaluation of the process conditions and the implementation of corrective actions regarding piping metallurgy and circuit design to prevent recurrence.

Incident Description and Background

On 2007-09-14, operations personnel identified a pinhole leak on a 90° elbow of 2-RV-102. This line is constructed of A106 Gr.B with an OD of 60.3 mm and a nominal thickness of 5.54 mm. The design limits for this circuit are 345 kPa(g) and 121 °C. Upon discovery, the line was isolated and a temporary leak repair clamp was installed to maintain unit integrity.

Initial review of the inspection history shows that LNDE-RPT-96-014 established the baseline thickness for 118 CMLs in 1996 before the unit entered service. The original QSCP-IP-ATU100 Rev 0 defined the inspection intervals and CML locations, classifying the regenerator overhead as Class 1 service. However, the CMLs were historically placed only on straight pipe runs, omitting fittings such as the failed elbow.

Analysis of Inspection Data

The most recent full-unit survey, HIS-2006-0412, was completed on 2006-02-24. While this survey adhered to the 5-year interval for Class 1 piping, it utilized spot UT readings at established CMLs rather than grid scans. Because the failed elbow lacked a designated CML, its localized thinning remained undetected during the 2006 survey. The site rule for remaining life calculations was applied only to the straight-run data, which did not reflect the accelerated loss at the fittings.

Following the leak, HIS-2007-0388 was commissioned to perform a comprehensive UT thickness survey and grid scans of every fitting in the regenerator overhead circuit. This report confirmed that while straight runs maintained thickness above the 2.79 mm t-min, multiple elbows exhibited localized thinning. The data from HIS-2007-0388 indicates that the corrosion was not uniform, rendering the previous spot-reading methodology insufficient for this circuit.

Root Cause Identification

The primary cause of the failure is localized corrosion resulting from condensing acid gas at the elbow. As described in API RP 571, the cooling of the overhead gas leads to the formation of aggressive condensate, which preferentially attacks changes in flow direction. The carbon steel metallurgy of 2-RV-102 provided insufficient resistance to this localized degradation mechanism under the operating environment of the ATU-100.

A significant contributing cause was the deficiency in the piping inspection plan. QSCP-IP-ATU100 Rev 0 failed to account for the higher turbulence and condensation rates at fittings, and the reliance on a 5-year interval for spot readings was inadequate for detecting localized phenomena. The absence of CMLs on elbows meant that the short-term corrosion rates calculated per API 570 did not represent the actual worst-case thinning occurring in the system.

Corrective Actions and Recommendations

Douglas Ferreira shall oversee the revision of the inspection plan to include mandatory CMLs on all fittings within the regenerator overhead circuit. Future inspections must utilize grid scanning rather than spot UT to ensure localized thinning is captured. The retirement thickness of 2.79 mm must be monitored using the larger of the long-term or short-term corrosion rates as established by the new baseline data in HIS-2007-0388.

Gordon Lisle will evaluate the feasibility of upgrading the elbow metallurgy or improving insulation to mitigate acid gas condensation. All Class 1 piping in the ATU-100 must be re-evaluated to ensure that CML placement complies with the

updated requirements for high-turbulence areas. The next inspection interval for 2-RV-102 will be set at half the remaining life or the 5-year Class 1 maximum, whichever is more frequent.

Sign-off	Name	Function	Date	Signature
Prepared	Douglas Ferreira	Inspection Supervisor, API 570 inspector	2007-11-30	_____
Prepared	Gordon Lisle	Unit Engineer ATU-100, P.Eng.	2007-11-30	_____